Math 629 Spring 26 **Name: Mengxiang Jiang**

Turn this assignment in on Gradescope. All work must be neat and legible. Turn in a "final draft". There should be nothing scratched out and your work should flow generally from left to right and top to bottom.

1. Show that $\sin x$ is continuous at all values of x (Hint: $\sin(x+\alpha)-\sin x = 2\sin(\frac{1}{2}\alpha)\cos(x+\frac{1}{2}\alpha)$).

   Using the identity in the hint, we take the absolute values:

   $$|\sin(x+\alpha)-\sin x| = 2\left|\sin\left(\frac{1}{2}\alpha\right)\right| \cdot \left|\cos\left(x+\frac{1}{2}\alpha\right)\right|.$$

   Since $|\cos x| \leq 1$ for all x, and $|\sin(x)| \leq |x|$ for all x, we have:

   $$|\sin(x+\alpha)-\sin x| \leq 2\left|\sin\left(\frac{1}{2}\alpha\right)\right| \leq 2\left|\frac{1}{2}\alpha\right| = |\alpha|.$$

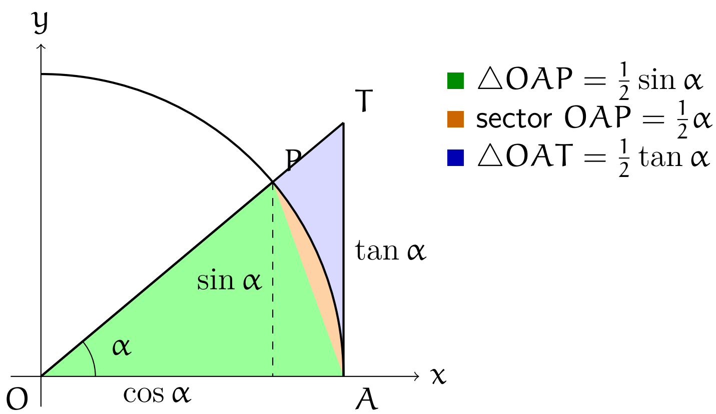
   Thus, for any $\epsilon > 0$, if we choose $\delta = \epsilon$, then whenever $|\alpha| < \delta$, we have:

   $$|\sin(x+\alpha)-\sin x| < \epsilon.$$

   This shows that $\sin x$ is continuous at all values of x.

2. What is $\lim_{a\to 0}\frac{\sin a}{a}$? Assuming that $\cos$ is continuous, use the above to find the derivative of $\sin x$.

   I remember getting points taken off an exam for using L'Hopital's for this, so I learned to do it through geometry. Consider the diagram below with three areas:

   From the diagram, we have the following inequalities:

   $$\frac{1}{2}\sin a \le \frac{1}{2}a \le \frac{1}{2}\tan a.$$

   Dividing through by $\frac{1}{2}\sin a$ and inverting the inequalities, we get:

   $$\cos a \le \frac{\sin a}{a} \le 1.$$

   Taking the limit as $a \to 0$, we have $\lim_{a\to 0}\cos a = 1$ and $\lim_{a\to 0} 1 = 1$. By the squeeze theorem, we conclude that $\lim_{a\to 0}\frac{\sin a}{a} = 1$.

   To find the derivative of $\sin x$, we use the definition of the derivative, the identity from the first question, and the limit we just found:

   $$\begin{aligned}\frac{d}{dx}\sin x &= \lim_{h\to 0}\frac{\sin(x+h)-\sin x}{h}\\ &= \lim_{h\to 0}\frac{2\sin\left(\frac{1}{2}h\right)\cos\left(x+\frac{1}{2}h\right)}{h}\\ &= \lim_{h\to 0}\cos\left(x+\frac{1}{2}h\right)\cdot\lim_{h\to 0}\frac{\sin\left(\frac{1}{2}h\right)}{\frac{1}{2}h}\\ &= \cos(x)\cdot 1 = \cos x.\end{aligned}$$

3. If $A < \frac{a_k}{b_k} < B$ for $k = 1, \dots, N$ then $A < \frac{\sum_{k=1}^{N} a_k}{\sum_{k=1}^{N} b_k} < B$.

   From $A < \frac{a_k}{b_k} < B$, we can multiply through by b_k to get:

$$Ab_k < a_k < Bb_k.$$

   Summing over k from 1 to N, we obtain:

$$A \sum_{k=1}^{N} b_k < \sum_{k=1}^{N} a_k < B \sum_{k=1}^{N} b_k.$$

   Dividing through by $\sum_{k=1}^{N} b_k$, we get:

$$A < \frac{\sum_{k=1}^{N} a_k}{\sum_{k=1}^{N} b_k} < B.$$